** Plan & Maps:-

Plan:-

→ A plan is the graphical representation of the features on the earth surface or below the earth surface as projected on a horizontal plane.

→ A plan is drawn on a large scale.

→ A plan is drawn for small area.

Map:-

→ The representation of earth surface on small scale, is called Map.

→ The map shows its geographical position on the globe.

→ on a map, Topography of the terrain, is depicted generally by Contours, hechures, and spot levels.

→ A map is drawn on small scale.

→ A map is drawn for large area.

** Unit of different Measurements:-

→ There are many units of measurement that are prevalent worldwide like the CGS Systems, FPS System, MKS Systems but the Standard one is the SI Systems.

→ past record of all survey works are usually in FPS System. Thus to use those records and any other records that are in different units, Those have to be Converted into SI Unit.

Length Unit Conversion.

1 inch = 0.0254 m. = 2.54 cm.
1 foot = 12 inch
1 yard = 3 feet
1 mile = 1760 yard = 5280 feet
1 Angstrom = 1×10^{-10} m.

Area Unit Conversion:-

1 square feet = 144 square inch
1 square yard (naj) = 9 sq ft. (square feet).
1 Acre = 4046.85 m^2
1 Hectare = 10^4 m^2
1 square feet = 0.0929 m^2.
1 Acre = 4840 sq yard.
1 hectare = 2.49 Acres approximately
= 10^4 m^2

⇒ Various measurement in surveying:-

→ In surveying the direction of gravity can always be taken as reference for all measurements.

→ The direction of gravity is established by suspending a plumb bob freely. This direction of gravity is taken as the vertical direction. Thus horizontal direction is at right angles to the vertical direction.

→ Any plane which Containing the horizontal line and perpendicular to the vertical direction is called as horizontal plane. The plane Containing a vertical line is called as vertical plane.

In Surveying, the following basic measurement are made.

a) Horizontal distance:-

→ The horizontal distance is measured in horizontal plane. on Sloping ground, the distance between two points is reduce to horizontal equivalent.

b) Horizontal angle:-

→ The horizontal angle is measured between two lines in horizontal plane. Theoretically the angle between two lines can vary from 0° to 360°.

c) Vertical distance:-

→ The direction of gravity is taken as vertical direction and thus vertical distance are measured in the direction of gravity. The vertical distance are measured to determine the difference of elevations between the various point.

d) vertical angle:-

→ vertical angle is measured between two lines in vertical plane.

⇒ Instruments used for various types of measurement:-

a). For horizontal measurement:- Tape, Chain, EDM, etc.

b). For horizontal angle measurement:-

→ Magnetic Compass, Theodolites, Total Station, Sextant etc.

c). For vertical distance:-

→ Tacheometer, levelling Instruments like dumpy level etc.

d). For vertical angle measurement:-

→ Sextant, Clinometer, theodolite etc.

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